

BITS, PILANI – K. K. BIRLA GOA CAMPUS

Theory of Computation

(CS F51)

Lecture No. 9

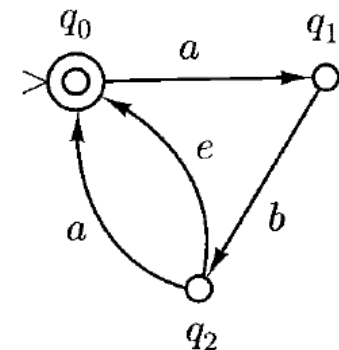
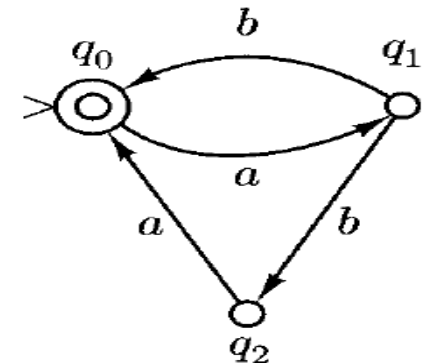
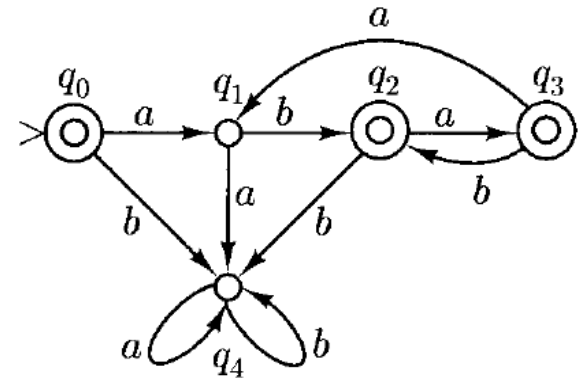


Nondeterministic Finite State Automaton (NFA)

Consider $L = (ab \cup aba)^*$

We will *relax* the following *constraints* of DFA:

1. We *may not define* transition function for each pair of state, input symbol combination.
2. We shall allow *several possible 'next states'* for a given combination of current state & input symbol
3. The machine allows *"empty transitions"* i.e. machine can move from one state to another state without reading the input



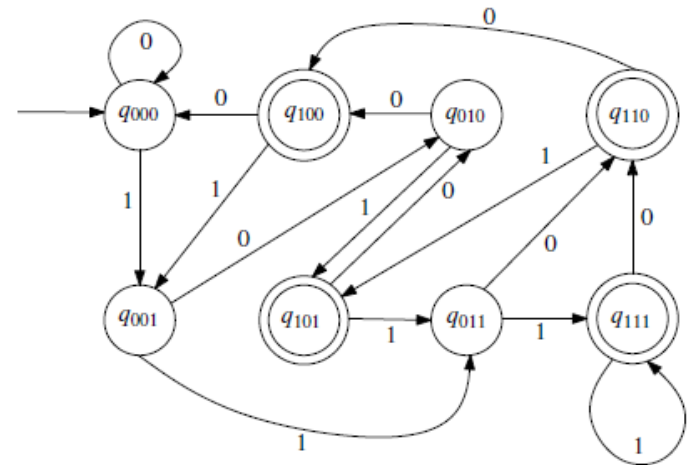
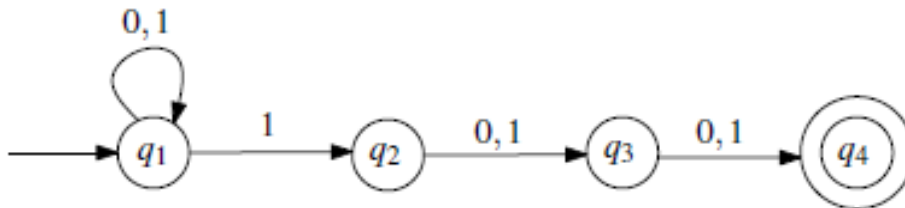
NFA - Example

Example: $L = \{w : \{0,1\}^* : w \text{ has a } 1 \text{ in the third position from the right}\}$

Idea

Strings of the type $x100$, $x101$, $x110$, $x111$ where $x \in \{0,1\}^*$ belongs to L

Remain in a state and when the input symbol is 1, *guess* that it is a string of the type above & then check for acceptance.



NFA

The automaton as it reads the input symbol, *may choose* at each step to go to any one of these *legal next states*.

The *choice* is *not determined by anything in our model* & is therefore said to be *nondeterministic*.

We think as NFA making the *right guess* for string acceptance.

Note:

NFA are *not realistic* model of computers.

Useful for *simplifying representation* of DFA

Important

Every NFA is *equivalent* to DFA.

Definition of NFA

Formal Definition

A *nondeterministic finite automaton* is a 5-tuple

$M = (K, \Sigma, \Delta, s, F)$, where

K is a finite set, whose elements are called *states*

Σ is a finite set, called the *alphabet* set

$s \in K$ is the *initial state*

$F \subseteq K$ is set of *final/accept* states

Δ , the transition relation is a subset of

$$K \times (\Sigma \cup \{e\}) \times K$$

NFA - Definitions

A triple of the form $(q, a, p) \in \Delta$, $a \in \Sigma \cup \{e\}$ is called a *transition* of M

This corresponds to machine is in state q and *the reading head* is pointing to a on input tape and moves to state p .

Configuration of NFA $(K, \Sigma, \Delta, s, F)$ is an element of $K \times \Sigma^*$

If (q, w) & (q', w') are two configurations of NFA M , then

$(q, w) \vdash_M (q', w')$ if and only if $w = aw'$ for some symbol $a \in \Sigma \cup \{e\}$ and $(q, a, q') \in \Delta$

Note that as in the case of DFA, $\vdash_M$ may not be *function*

As before, we denote the *reflexive, transitive closure* of

$\vdash_M$ by $\vdash_M^*$

NFA - Definitions

A string $w \in \Sigma^*$ is said to be *accepted* by NFA M if and only if there is a state $q \in F$ (set of final states) such that $(s, w) \vdash_M^* (q, e)$ where s is the initial state.

Note:

String is *accepted* if there is *some sequence of moves* that leads NFA from an initial configuration to a configuration in which the state component is a final state.

The *language accepted* by the NFA M , $L(M)$ is the set of all strings accepted by M .

Note: A string is *rejected* by M if there is *no sequence of moves* that leads NFA from an initial configuration to a configuration in which the state component is a final state.

Main Theorem

Two finite automata M_1 & M_2 are *equivalent* if and only if $L(M_1) = L(M_2)$

It means the two automata's generate the same language may be *using different methods*.

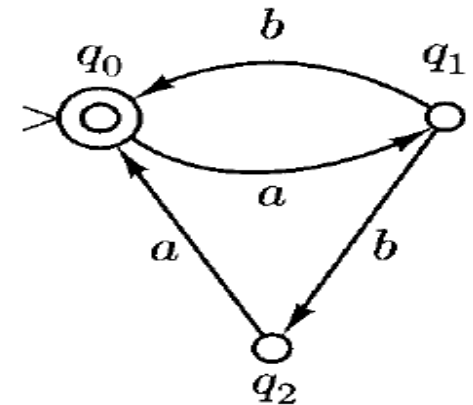
Theorem:

For each NFA, there is an equivalent DFA.

Proof: Key Idea

For pair (q_1, b) *think* of NFA as occupying *set of states* $\{q_0, q_2\}$

rather than a single undetermined member of that set.



The *construction* formalizes this idea.

Proof Continued

Let $M = (K, \Sigma, \Delta, s, F)$ be a *NFA*.

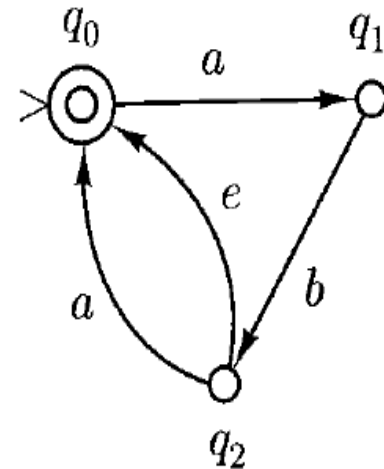
To construct $M' = (K', \Sigma, \delta, s', F')$ *DFA* such that
 $L(M) = L(M')$

The *set of states of M'* will be $K' = 2^K$ all subsets of K

The *set of final states of M'* will consists of all those subsets of K that contains *at least one final state* of M .

Transition function for M'

A *move* of M' on reading $a \in \Sigma$ *imitates* a move of M on input symbol a & *possibly* followed by any number of *e-moves* of M .



Proof Continued

Definition:

For any state $q \in K$, $E(q)$ denote the set of all states of M that are *reachable from state q* without reading any input.

So $E(q) = \{p \in K : (q, e) \vdash_M^* (p, e)\}$

Formal definition of DFA $M' = (K', \Sigma, \delta, s', F')$

$$K' = 2^K$$

$$s' = E(s)$$

Easy to see that M' is an DFA

$$F' = \{Q \subseteq K : Q \cap F \neq \emptyset\}$$

Remains to prove that
 $L(M) = L(M')$

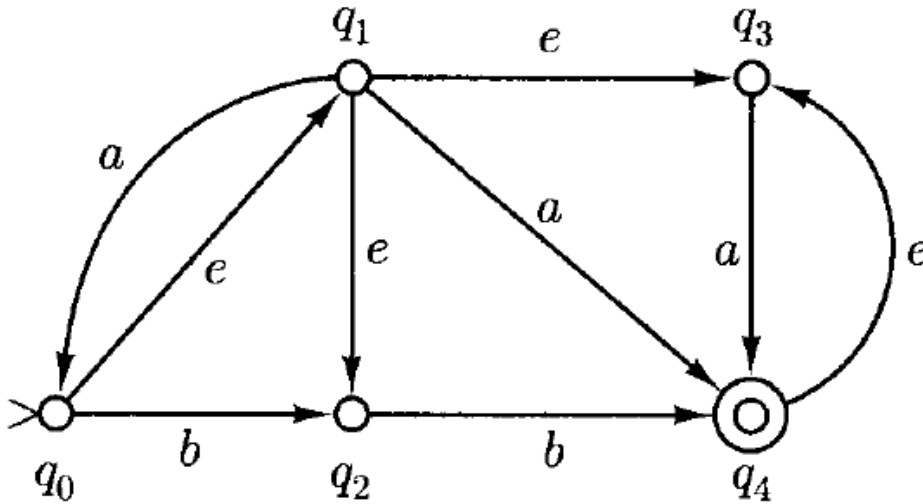
For each $Q \subseteq K$ & each $a \in \Sigma$, define

$$\delta(Q, a) = \cup \{E(p) : p \in K \text{ \& } (q, a, p) \in \Delta \text{ for some } q \in Q\}$$

$\delta(Q, a)$ is set of all states on M to which M can go by *reading* input a & possibly following *several e transitions*.

Proof Continued

Example

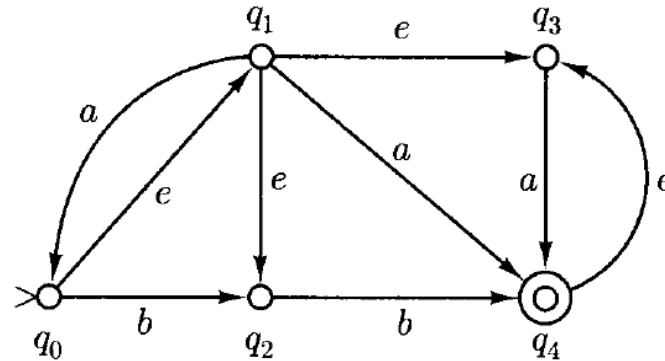


Then $s' = E(s) = \{q_0, q_1, q_2, q_3\}$

NFA to DFA Construction

Example

Consider NFA M



Note

Since M has 5 states, corresponding DFA M' will have $2^5 = 32$ states.

However, in practice only a few of these states will be *reachable* from initial state s' by reading some input string. We shall build this *reachable* part of M' by starting from s' & introducing a new state only when it is needed.

NFA to DFA Construction

$$s' = E(q_0)$$

$$\delta(Q, a) = \cup \{E(p) : p \in K \text{ \& } (q, a, p) \in \Delta \text{ for some } q \in Q\}$$

$$= \{q_0, q_1, q_2, q_3\}$$

$$(q_1, a, q_0), (q_1, a, q_4)$$

$$\& (q_3, a, q_4)$$

Are all the transitions

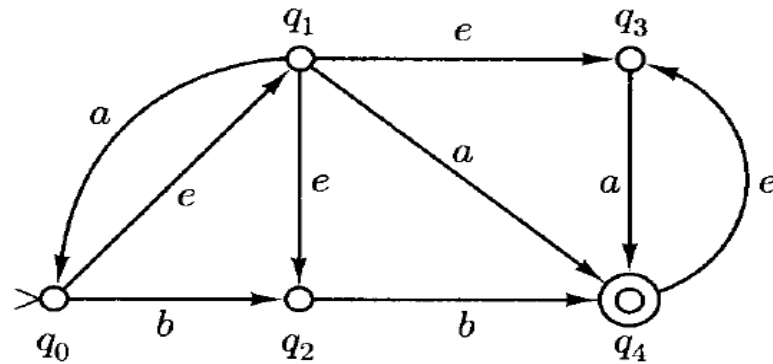
(q, a, p) for some $q \in s'$

$$\therefore \delta(s', a) = E(q_0) \cup E(q_4) = \{q_0, q_1, q_2, q_3, q_4\}$$

Similarly, $(q_0, b, q_2), (q_2, a, q_4)$

Are all the transitions (q, b, p) for some $q \in s'$

$$\therefore \delta(s', b) = E(q_2) \cup E(q_4) = \{q_2, q_3, q_4\}$$



NFA to DFA Construction

The *new states* introduced

Are $\{q_0, q_1, q_2, q_3, q_4\}$ and
 $\{q_2, q_3, q_4\}$

We *repeat* the construction
the newly introduced states

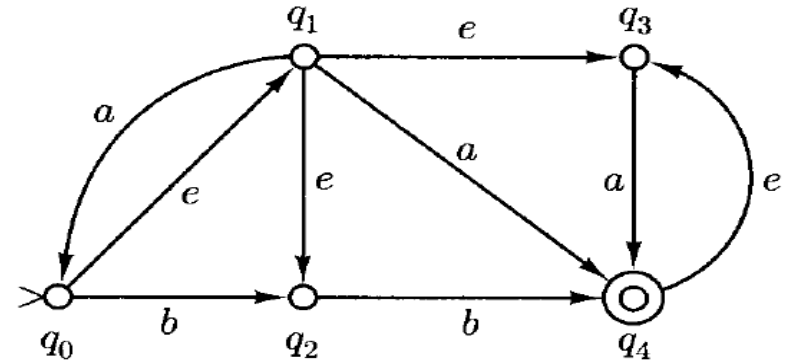
$$\delta(\{q_0, q_1, q_2, q_3, q_4\}, a) = E(q_0) \cup E(q_4) = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\delta(\{q_0, q_1, q_2, q_3, q_4\}, b) = E(q_2) \cup E(q_4) = \{q_2, q_3, q_4\}$$

$$\delta(\{q_2, q_3, q_4\}, a) = E(q_4) = \{q_3, q_4\}$$

$$\delta(\{q_2, q_3, q_4\}, b) = E(q_4) = \{q_3, q_4\}$$

The *new state* introduced is $\{q_3, q_4\}$



NFA to DFA Construction

Next

$$\delta(\{q_3, q_4\}, a) = E(q_4) = \{q_3, q_4\}$$

$$\delta(\{q_3, q_4\}, b) = \phi$$

The *new state* introduced is ϕ

And finally,

$$\delta(\phi, a) = \delta(\phi, b) = \phi$$

So *no new state* is discovered & we *terminate* the process.

F' the set of *final states*

contains each set of states of which q_4 is a member.

